

Tutorial Sheet 2

Dr G. Papadakis, g.papadakis@ic.ac.uk, Room CABG 331

1. The velocity potential of a flow is given by

$$\phi = \frac{U_0(x^2 - y^2)}{2l}$$

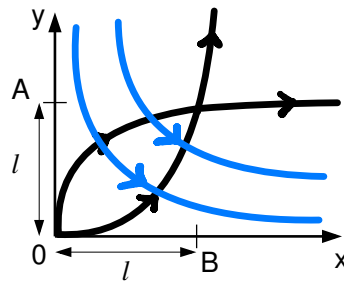
(a) Calculate the two velocity components and the stream function ψ . Hence show that:

(i) this is irrotational flow;

(ii) this is the flow in the right-angled corner shown below (sketch the streamlines);

(iii) the speed is U_0 at the points A ($x = 0, y = l$) and B ($x = l, y = 0$) on the walls.

(b) Calculate the pressure on the walls in terms of p_0 , the pressure at the corner, 0. What is the total force (per unit depth) exerted by the fluid on the segment AOB of the wall and its direction, if the pressure on the other side of the wall is p_0 ?



$$(a) \quad u = \frac{\partial \phi}{\partial x} = \frac{U_0 x}{l} \quad , \quad v = \frac{\partial \phi}{\partial y} = -\frac{U_0 y}{l}$$

$$\therefore \psi_x = \int u \, dy = \frac{U_0 xy}{l} + h(x)$$

$$\psi_y = \int -v \, dx = \frac{U_0 xy}{l} + h(y)$$

$$\therefore \psi = \frac{U_0 xy}{l}$$

$$(i) \omega = \nabla \times u = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 - 0 = 0$$

(ii) Show in the graph above

$$(i) \text{ At } A(0, l), V = \sqrt{u^2 + v^2} = U_0$$

$$\text{At } B(l, 0), V = \sqrt{u^2 + v^2} = U_0$$

$$(b) p_{\text{corner}} = 0$$

$$\therefore p_0 = p_s + \frac{1}{2} \rho V^2 = p_s + \frac{1}{2} \rho (u^2 + v^2) = p_s + \frac{1}{2} \rho \frac{U_0^2}{l^2} (x^2 + y^2)$$

$$\therefore p_{\text{wall}} = p_s = p_0 - \frac{1}{2} \rho \frac{U_0^2}{l^2} (x^2 + y^2)$$

$$\therefore p_{\text{wall}} = p_s - \frac{1}{2} \rho \frac{U_0^2}{l^2} x^2 \text{ or } p_0 - \frac{1}{2} \rho \frac{U_0^2}{l^2} y^2 \text{ at wall}$$

$$\therefore \frac{F}{d} = \frac{pA}{d} = pL = \int_L p \, dL = \sqrt{\left(\int_x p \, dx\right)^2 + \left(\int_y p \, dy\right)^2}$$

$$= \sqrt{\left(\int_0^l \left(p_s - \frac{1}{2} \rho \frac{U_0^2}{l^2} x^2 - p_0\right) dx\right)^2 + \left(\int_0^l \left(p_s - \frac{1}{2} \rho \frac{U_0^2}{l^2} y^2 - p_0\right) dy\right)^2}$$

$$= \sqrt{\left(\left[-\frac{1}{6} \rho \frac{U_0^2}{l^2} x^3\right]_0^l\right)^2 + \left(\left[-\frac{1}{6} \rho \frac{U_0^2}{l^2} y^3\right]_0^l\right)^2}$$

$$= \sqrt{\frac{1}{36} \rho^2 U_0^4 l^4 + \frac{1}{36} \rho^2 U_0^4 l^4} = \frac{\sqrt{2}}{6} \rho U_0^2$$

at incidence 45°

2. The streamfunction

$$\psi = U_{\infty}y + k \left(\tan^{-1} \left(\frac{y}{x+x_0} \right) - \tan^{-1} \left(\frac{y}{x-x_0} \right) \right)$$

corresponds to flow round an oval shaped body. By calculating the velocity components or otherwise show that the flow is irrotational. By locating the stagnation points on the axis find the chord length of the body.

for $y = \tan^{-1} x = \arctan x$, $x = \tan y$

$$\therefore \frac{dx}{dy} = \sec^2 y = \frac{1}{\cos^2 y} = \frac{\cos^2 y + \sin^2 y}{\cos^2 y} = 1 + \tan^2 y = 1 + x^2$$

$$\therefore \frac{dy}{dx} = \frac{1}{1+x^2}$$

for ψ

$$\begin{aligned} u = \frac{\partial \psi}{\partial y} &= U_{\infty} + k \left[(x+x_0)^{-1} \left(1 + \left(\frac{y}{x+x_0} \right)^2 \right)^{-1} - (x-x_0)^{-1} \left(1 + \left(\frac{y}{x-x_0} \right)^2 \right)^{-1} \right] \\ &= U_{\infty} + k \left[\frac{x+x_0}{(x+x_0)^2 + y^2} - \frac{x-x_0}{(x-x_0)^2 + y^2} \right] \end{aligned}$$

$$\begin{aligned} v = -\frac{\partial \psi}{\partial x} &= -k \left[-y(x+x_0)^{-2} \left(1 + \left(\frac{y}{x+x_0} \right)^2 \right)^{-1} + y(x-x_0)^{-2} \left(1 + \left(\frac{y}{x-x_0} \right)^2 \right)^{-1} \right] \\ &= k \left[y(x+x_0)^{-2} \left(1 + \left(\frac{y}{x+x_0} \right)^2 \right)^{-1} - y(x-x_0)^{-2} \left(1 + \left(\frac{y}{x-x_0} \right)^2 \right)^{-1} \right] \\ &= k \left[\frac{y}{(x+x_0)^2 + y^2} - \frac{y}{(x-x_0)^2 + y^2} \right] \end{aligned}$$

$$\therefore \frac{\partial u}{\partial y} = k \left[\frac{-2y(x+x_0)}{(x+x_0)^2 + y^2} - \frac{-2y(x-x_0)}{(x-x_0)^2 + y^2} \right]$$

$$\frac{\partial v}{\partial x} = k \left[\frac{-y[2(x+x_0)]}{(x+x_0)^2 + y^2} - \frac{-y[2(x-x_0)]}{(x-x_0)^2 + y^2} \right]$$

$$\therefore \omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

At stagnation point, $u = v = 0$

$$\therefore \text{for } v = 0, \frac{y}{(x+x_0)^2 + y^2} = \frac{y}{(x-x_0)^2 + y^2}$$

$$\therefore y = 0 \text{ or } x = 0$$

$$\therefore \text{for } u = 0, U_\infty + k \left[\frac{x+x_0}{(x+x_0)^2 + y^2} - \frac{x-x_0}{(x-x_0)^2 + y^2} \right] = 0$$

$$\therefore k \left[\frac{x-x_0}{(x-x_0)^2 + y^2} - \frac{x+x_0}{(x+x_0)^2 + y^2} \right] = U_\infty$$

$\therefore$ Only valid for $y = 0$

$$\therefore \frac{U_\infty}{k} = \frac{1}{x+x_0} - \frac{1}{x-x_0} = \frac{-2x_0}{x^2 - x_0^2}$$

$$\therefore x = \pm \sqrt{x_0^2 + \frac{2x_0 R}{U_\infty}} \quad \therefore C = |x_1 - x_0| = 2\sqrt{x_0^2 + \frac{2x_0 R}{U_\infty}}$$

3. (a) Use the complex potential for horizontal flow U_∞ about a circular cylinder with circulation Γ to calculate the u -component of velocity on the cylinder surface as a function of θ , the angular coordinate.
- (b) Hence show that the positions of the front and rear stagnation points on the cylinder are given by:

$$\theta = \sin^{-1} \left(\frac{\Gamma}{4\pi R U_\infty} \right)$$

- (c) What magnitude of Γ causes these two points to coincide on the surface and where? Sketch the flow field for larger magnitudes of Γ .

$$\begin{aligned} \text{(a)} \quad F(z) &= U_\infty \left(z + \frac{R^2}{z} \right) - \frac{i\Gamma}{2\pi} \ln z \\ &= U_\infty \left(r e^{i\theta} + \frac{R^2}{r} e^{-i\theta} \right) - \frac{i\Gamma}{2\pi} (\ln r + i\theta) \\ &= U_\infty \left[r (\cos\theta + i\sin\theta) + \frac{R^2}{r} (\cos\theta - i\sin\theta) \right] \\ &\quad - \frac{i\Gamma}{2\pi} (\ln r + i\theta) \end{aligned}$$

$$\therefore \phi = \operatorname{Re}(F) = U_\infty r \cos\theta + U_\infty \frac{R^2}{r} \cos\theta + \frac{\Gamma}{2\pi} \theta$$

$$\psi = \operatorname{Im}(F) = U_\infty r \sin\theta - U_\infty \frac{R^2}{r} \sin\theta - \frac{\Gamma}{2\pi} \ln r$$

$$\therefore V_\theta = -\frac{\partial\psi}{\partial r} = -U_\infty \sin\theta - U_\infty \frac{R^2}{r^2} \sin\theta + \frac{\Gamma}{2\pi r}$$

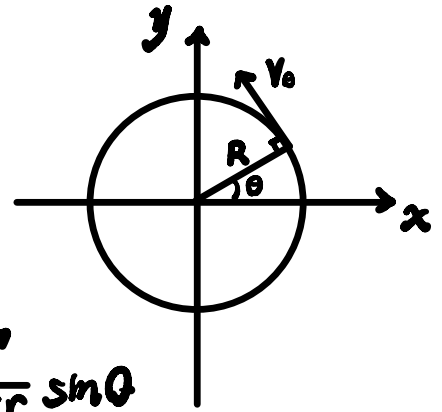
$$V_r = \frac{\partial\phi}{\partial r} = U_\infty \cos\theta - U_\infty \frac{R^2}{r^2} \cos\theta$$

At cylinder surface, $r = R$

$$\therefore V_\theta = -2U_\infty \sin\theta + \frac{\Gamma}{2\pi r}$$

$$V_r = 0$$

$$\therefore u = V_\theta \sin\theta = -2U_\infty \sin^2\theta + \frac{\Gamma}{2\pi r} \sin\theta$$



(b) At stagnation point, $V_\theta = 0$

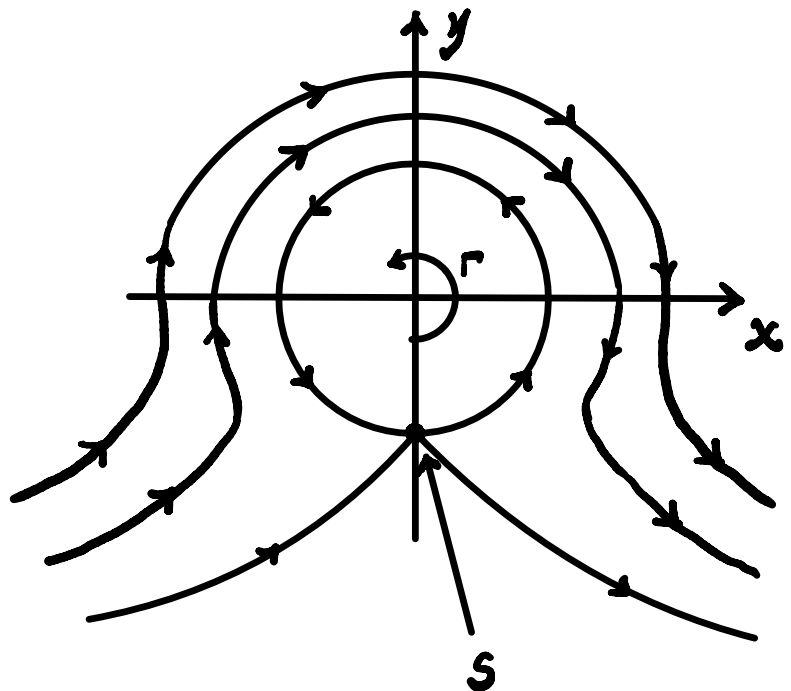
$$\therefore -2U_\infty \sin\theta + \frac{\Gamma}{2\pi R} = 0$$

$$\therefore \theta = \sin^{-1}\left(\frac{\Gamma}{4\pi U_\infty R}\right)$$

(c) if there is only one solution for $\theta = \sin^{-1}\left(\frac{\Gamma}{4\pi U_\infty R}\right)$

$$\frac{\Gamma}{4\pi U_\infty R} = 1$$

$$\therefore \Gamma = 4\pi U_\infty R$$

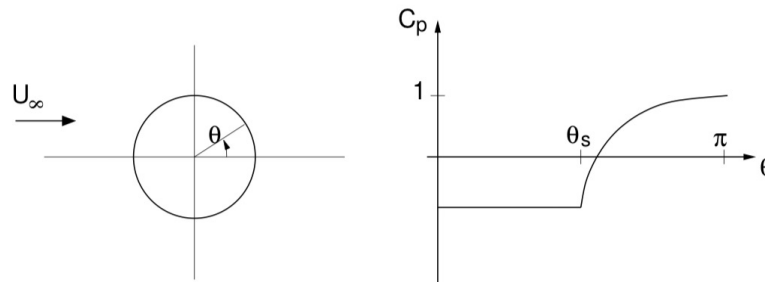


4. An attempt to model real flow about a circular cylinder might be as follows:

For the front part of the cylinder up to the separation point $\theta = \theta_s$, the surface velocity and surface pressure (see following figure) to be given by potential flow:

$$\phi = U_{\infty} \left(r + \frac{R^2}{r} \right) \cos \theta$$

For the rest of the surface, the pressure to be assumed to be constant equal to the value given by the potential flow at $\theta = \theta_s$, as shown in the figure.



Taking $\theta_s = 99^\circ$ (equivalent to separation 81° from the front stagnation point), calculate the drag coefficient for the cylinder, assuming that the pressure distribution is symmetric about the $\theta = 0$, i.e. x-axis. Comment on the result.

$$V_r = \frac{\partial \phi}{\partial r} = U_{\infty} \left(1 - \frac{R^2}{r^2} \right) \cos \theta$$

$$V_{\theta} = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = -U_{\infty} \left(1 + \frac{R^2}{r^2} \right) \sin \theta$$

At surface of circle, $r = R$

$$V_r = 0, \quad V_{\theta} = -2U_{\infty} \sin \theta$$

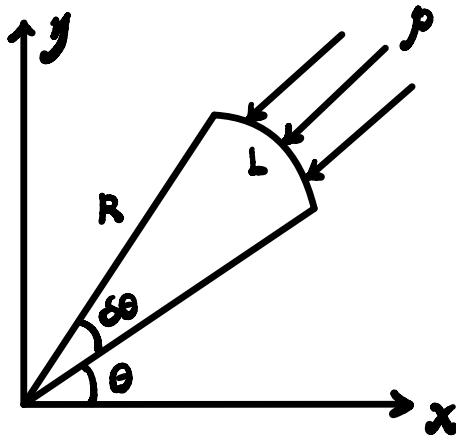
$$\therefore C_p = \frac{p - p_{\infty}}{\frac{1}{2} \rho U_{\infty}^2}$$

According to Bernoulli, $p_0 = p + \frac{1}{2} \rho V_{\theta}^2 = p_{\infty} + \frac{1}{2} \rho U_{\infty}^2$

$$\therefore C_p = \frac{\frac{1}{2} \rho U_{\infty}^2 - \frac{1}{2} \rho V_{\theta}^2}{\frac{1}{2} \rho U_{\infty}^2} = 1 - \left(\frac{V_{\theta}}{U_{\infty}} \right)^2 = 1 - 4 \sin^2 \theta$$

$$\therefore C_p = \begin{cases} 1 - 4\sin^2(99^\circ) & (0 \leq \theta \leq 99^\circ \text{ and } 261^\circ \leq \theta \leq 360^\circ) \\ 1 - 4\sin^2\theta & (99^\circ \leq \theta \leq 261^\circ) \end{cases}$$

Applying conservation of momentum in C.V. with $V_r = 0$, $F_p + D = 0$



$$D_{x-o} = -p_h L$$

$$= -p_h R \delta\theta$$

$$= -\int p \cos\theta R d\theta$$

$$\therefore C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}$$

$$\therefore p = \frac{1}{2}\rho U_\infty^2 C_p + p_\infty$$

$$\therefore D_{x-o} = -\int \frac{1}{2}\rho U_\infty^2 C_p \cos\theta R d\theta - \int p_\infty \cos\theta R d\theta$$

$$= -\frac{1}{2}\rho U_\infty^2 R \int C_p \cos\theta d\theta - p_\infty R \int \cos\theta d\theta$$

$$\therefore C_D = \frac{D_{x-o}}{\frac{1}{2}\rho U_\infty^2 C} = \frac{-\frac{1}{2}\rho U_\infty^2 R \int C_p \cos\theta d\theta - p_\infty R \int \cos\theta d\theta}{\frac{1}{2}\rho U_\infty^2 (2R)}$$

$$= -\frac{1}{2} \int C_p \cos\theta d\theta - \frac{p_\infty \int_0^{180} \cos\theta d\theta}{\rho U_\infty^2}$$

$$= -\int_0^{99} (1 - 4\sin^2(99^\circ)) \cos\theta d\theta - \int_{99}^{180} (1 - 4\sin^2\theta) \cos\theta d\theta - 0$$

$$\begin{aligned}
 \therefore C_D &= - \left[(1 - 4 \sin^4 44^\circ) \sin \theta \right]_0^{99} - \left[\sin \theta \right]_{99}^{180} \\
 &\quad + \left[\frac{4}{3} \sin^3 \theta \right]_{99}^{180} \\
 &= 2.5694
 \end{aligned}$$

Drag coefficient of circular cylinder is very large

5. For a Rankine oval arising from the superposition of a uniform stream at zero incidence, U_∞ , a source m at location $(-a, 0)$ and a sink $-m$ at location $(a, 0)$, show that the half-length, l , and half-height, h , of the Rankine-oval are given by

$$\frac{l}{a} = \left(\frac{m}{\pi U_\infty a} + 1 \right)^{\frac{1}{2}} \quad \text{and} \quad \frac{h}{a} = \frac{m}{\pi U_\infty a} \cot^{-1} \left(\frac{h}{a} \right)$$

$$F(z) = U_\infty z + \frac{m}{2\pi} \ln(z+a) - \frac{m}{2\pi} \ln(z-a)$$

$$\text{Let } z_1 = z+a = r_1 e^{i\theta_1},$$

$$z_2 = z-a = r_2 e^{i\theta_2}$$

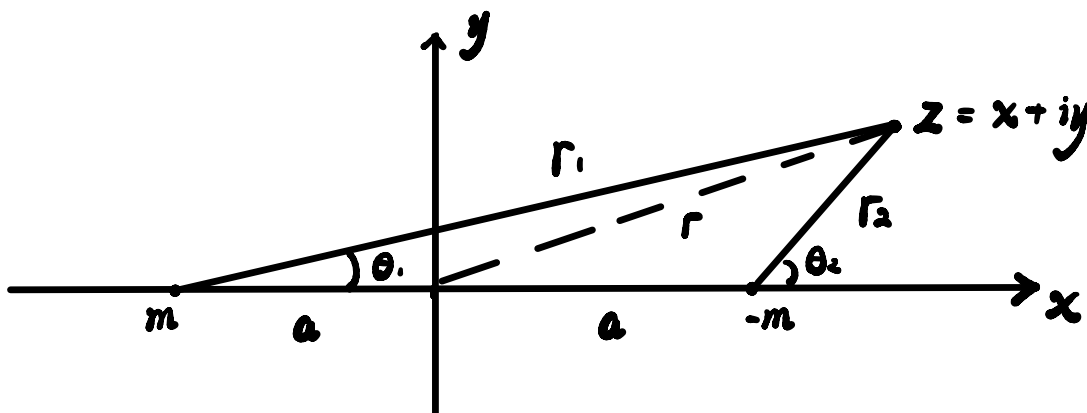
$$\therefore F(z) = U_\infty z + \frac{m}{2\pi} \ln(z_1) - \frac{m}{2\pi} \ln(z_2)$$

$$= U_\infty r e^{i\theta} + \frac{m}{2\pi} \ln(r_1) + i \frac{m}{2\pi} \theta_1 - \frac{m}{2\pi} \ln(r_2) - i \frac{m}{2\pi} \theta_2$$

$$= U_\infty (x+iy) + \frac{m}{2\pi} \ln\left(\frac{r_1}{r_2}\right) + i \frac{m}{2\pi} (\theta_1 - \theta_2)$$

$$\therefore \phi = U_\infty x + \frac{m}{2\pi} \ln\left(\frac{r_1}{r_2}\right)$$

$$\psi = U_\infty y + \frac{m}{2\pi} (\theta_1 - \theta_2)$$



$$\therefore \theta_1 = \tan^{-1} \left(\frac{y}{x+a} \right)$$

$$\theta_2 = \tan^{-1} \left(\frac{y}{x-a} \right)$$

$$\therefore \psi = U_\infty y + \frac{m}{2\pi} \left[\tan^{-1} \left(\frac{y}{x+a} \right) - \tan^{-1} \left(\frac{y}{x-a} \right) \right]$$

for $y = \tan^{-1} x = \arctan x$, $x = \tan y$

$$\therefore \frac{dx}{dy} = \sec^2 y = \frac{1}{\cos^2 y} = \frac{\cos^2 y + \sin^2 y}{\cos^2 y} = 1 + \tan^2 y = 1 + x^2$$

$$\therefore \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\therefore u = \frac{\partial \psi}{\partial y}$$

$$= U_\infty + \frac{m}{2\pi} \left[\frac{1}{x+a} \frac{1}{1 + \left(\frac{y}{x+a} \right)^2} - \frac{1}{x-a} \frac{1}{1 + \left(\frac{y}{x-a} \right)^2} \right]$$

$$= U_\infty + \frac{m}{2\pi} \left(\frac{x+a}{(x+a)^2 + y^2} - \frac{x-a}{(x-a)^2 + y^2} \right)$$

$$v = - \frac{\partial \psi}{\partial x}$$

$$= 0 + \frac{m}{2\pi} \left[\frac{y}{(x+a)^2 + y^2} - \frac{y}{(x-a)^2 + y^2} \right]$$

$$= \frac{m}{2\pi} \left(\frac{y}{(x+a)^2 + y^2} - \frac{y}{(x-a)^2 + y^2} \right)$$

When $\gamma = 0$,

$$y = 0 \quad \text{or} \quad x = 0$$

At stagnation points, $y = 0$ and $u = 0$

$$\therefore U_{\infty} + \frac{m}{2\pi} \left(\frac{x+a}{(x+a)^2 + y^2} - \frac{x-a}{(x-a)^2 + y^2} \right) = 0$$

$$\therefore \frac{2\pi U_{\infty}}{m} = \frac{1}{x-a} - \frac{1}{x+a} = \frac{2a}{x^2 - a^2}$$

$$\therefore x^2 = a^2 + \frac{ma}{\pi U_{\infty}} \quad \therefore x = \pm \sqrt{a^2 + \frac{ma}{\pi U_{\infty}}}$$

At points where $|U|$ is maximum, $x = 0$, $y = h$

$$\therefore \psi = U_{\infty} y + \frac{m}{2\pi} \left[\tan^{-1} \left(\frac{y}{x+a} \right) - \tan^{-1} \left(\frac{y}{x-a} \right) \right] = 0$$

$$\therefore U_{\infty} h + \frac{m}{2\pi} \left[\tan^{-1} \left(\frac{h}{a} \right) - \tan^{-1} \left(-\frac{h}{a} \right) \right] = 0$$

$$\therefore U_{\infty} h + \frac{m}{\pi} \tan^{-1} \left(\frac{h}{a} \right) = 0 \quad \therefore U_{\infty} h = -\frac{m}{\pi} \tan^{-1} \left(\frac{h}{a} \right) = \frac{m}{\pi} \cot^{-1} \left(\frac{h}{a} \right)$$

$$\therefore \frac{b}{a} = \frac{|x|}{a} = \left(1 + \frac{m}{\pi U_{\infty} a} \right)^{\frac{1}{2}}$$

$$\frac{h}{a} = \frac{m}{U_{\infty} \pi a} \cot^{-1} \left(\frac{h}{a} \right)$$

6. A crude approximation to a stationary hurricane is to model the flow-field outside its core by the 2-D flow due to a the superposition of a free vortex and a sink.

- Calculate the angle that the resulting velocity makes with the inward radial direction.
- Sketch the streamlines.
- If the wind velocity 20km from the centre of the hurricane is at an angle of 65° with the inward-radial direction, and the pressure at this distance is 550 N/m^2 less than the pressure far away, calculate the strength of the apparent sink (i.e. the rate of volume influx of air per metre height) and the circulation.

$$\begin{aligned}
 (Q) \quad F(z) &= -\frac{i\Gamma}{2\pi} \ln z - \frac{m}{2\pi} \ln z = -\left(\frac{i\Gamma}{2\pi} + \frac{m}{2\pi}\right) \ln z \\
 &= -\left(\frac{i\Gamma}{2\pi} + \frac{m}{2\pi}\right) \ln r e^{i\theta} = -\left(\frac{i\Gamma}{2\pi} + \frac{m}{2\pi}\right) (\ln r + i\theta) \\
 &= \left(-\frac{m}{2\pi} \ln r + \frac{\Gamma}{2\pi} \theta\right) + \left(-\frac{\Gamma}{2\pi} \ln r - \frac{m}{2\pi} \theta\right) i
 \end{aligned}$$

$$\therefore \phi = -\frac{m}{2\pi} \ln r + \frac{\Gamma}{2\pi} \theta$$

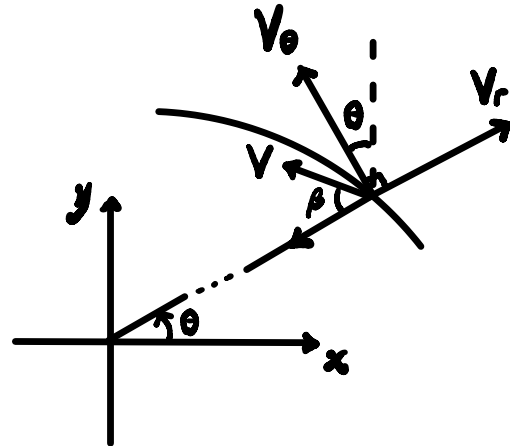
$$\psi = -\frac{\Gamma}{2\pi} \ln r - \frac{m}{2\pi} \theta$$

$$\therefore V_r = \frac{d\phi}{dr} = -\frac{m}{2\pi r}$$

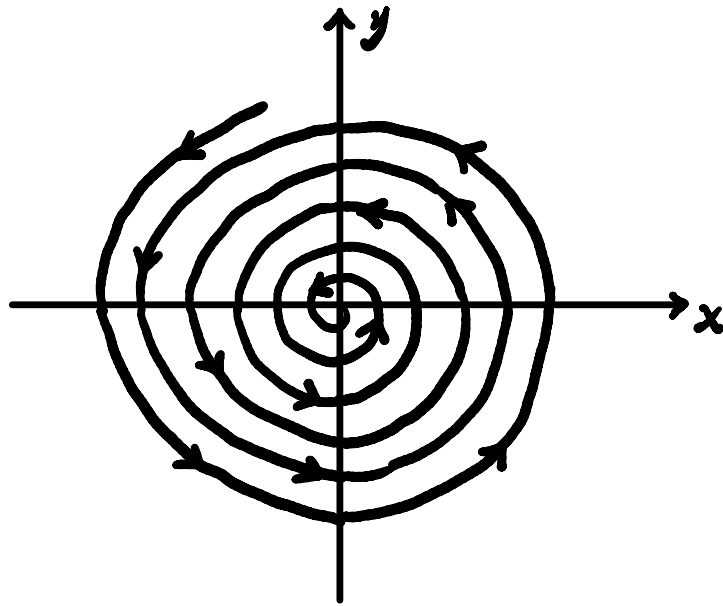
$$V_\theta = -\frac{d\psi}{dr} = \frac{\Gamma}{2\pi r}$$

$$\therefore \tan \beta = \frac{V_\theta}{-V_r}$$

$$\therefore \beta = \tan^{-1} \left(\frac{V_\theta}{-V_r} \right) = \tan^{-1} \left(\frac{\Gamma}{m} \right)$$



(b)



(c) When $\beta = 65^\circ$, $\frac{\Gamma}{m} = \tan(65^\circ)$

$$\therefore p_0 = p_s + \frac{1}{2}\rho U^2$$

as the condition along the streamline is the same

$$\therefore U = \left(\frac{2(p_0 - p_s)}{\rho} \right)^{\frac{1}{2}} = \left(\frac{1100}{\rho} \right)^{\frac{1}{2}} = \frac{20}{7} \sqrt{110} \text{ (m/s)}$$

$$\therefore V_\theta = \frac{\Gamma}{2\pi r} = \sin\beta U = 27.158 \text{ (m/s)}$$

$$\therefore \Gamma = 3.413 \times 10^6 \text{ (m}^2\text{/s)}$$

$$m = \frac{\Gamma}{\tan(65^\circ)} = 1.591 \times 10^6 \text{ (m}^2\text{/s)}$$

Numerical answers:

1. (a) $u = U_0 x/l$, $v = -U_0 y/l$, $\psi = U_0 xy/l + \text{const}$ (b) $p = p_0 - \frac{1}{2}\rho U_0^2 y^2/l^2$; force $= \frac{\sqrt{2}\rho U_0^2 l}{6}$
2. $2(x_0^2 + \frac{2kx_0}{U_\infty})^{\frac{1}{2}}$ 3. (c) $|\Gamma| = 4\pi R U_\infty$ 4. $C_D = 2.57$
5. Hint for h/a : $\tan(\pi/2 - X) = \cot(X)$
6. (a) $\tan^{-1}(\Gamma/m)$ (c) $m = 1.59 \times 10^6 \text{ m}^2/\text{s}$, $\Gamma = 3.41 \times 10^6 \text{ m}^2/\text{s}$